

Joint Robust Beamforming Design for WPT-Assisted D2D Communications in MISO-NOMA: Fractional Programming and Deep Reinforcement Learning

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Abstract: This paper introduces a framework aimed at aiding the development of sixth-generation (6G) ultra-massive machine type communications (umMTC). Precisely, the deployment of wireless power transfer (WPT) supported device-to-device (D2D) communication occurs within multiple-input single-output non-orthogonal multiple access (MISO-NOMA) downlink networks to facilitate spectrum and energy collaboration. A pure fractional programming (PFP) algorithm is proposed to maximize the WPT-assisted device's energy efficiency. An optimal closed-form solution for determining the time-switching coefficient of the WPT device is provided. For the robust beamforming design, the complex multi-dimension quadratic transform is applied. Moreover, the paper applies the deep deterministic policy gradient (DDPG)-based approach to directly address the problem and compares it with the proposed algorithm. Simulation outcomes highlight two key insights: 1) The PFP algorithm surpasses the performance of the DRL-based algorithm when the acquired channel state information (CSI) is accurate

or contains negligible errors, while the opposite is true for imperfect CSI 2) The higher energy efficiency gains can be achieved in NOMA scheme than that in Orthogonal Multiple Access (OMA) scheme.

Keywords: DRL; D2D; fractional-programming; NOMA; WPT

I. INTRODUCTION

The progression from fifth-generation (5G) to sixth-generation (6G) in wireless communication has amplified the requirements from massive machine-type communications (mMTC) to a broader scale, known as ultra-mMTC (umMTC) [1]. This surge is largely attributed to the advent of new applications and usage scenarios, such as the internet of everything (IoE). However, the development of a dedicated communication system for umMTC presents a substantial, time-intensive, and costly endeavor. On the other hand, accessing more active communication devices in the conventional orthogonal manner, spectrum constraints and energy consumption will be exacerbated. To counteract these challenges, this paper introduces a novel approach by integrating wireless power transfer (WPT)-assisted device-to-device (D2D) communi-

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cation with non-orthogonal multiple access (NOMA) technology, thereby offering a solution that enables efficient spectrum and energy cooperation.

From the spectrum perspective, NOMA has been regarded as a promising solution to support mMTC in the future 6G networks due to its spectrum sharing mechanism among wireless devices [2–6], where successive interference cancellation (SIC) has been utilized to remove mutual interferences [7]. In [8], it has been demonstrated that the spectral efficiency of IoT networks can be significantly improved if NOMA is applied. In addition, for the design of ultra-dense networks, a significant number of D2D devices will be integrated into the cellular infrastructure to enable emerging applications, such as healthcare, smart home and so on [9–11]. The key feature of D2D communications is a kind of short-distance peer-to-peer wireless communication that enables linkage between devices while eliminating the involvement of access points (APs) and base stations (BSs), and hence can meet the demands of massive connections without augmenting the load on BSs [12–13]. To conventional orthogonal multiple access (OMA) based protocols, any permitted access device is allowed to solely occupy a resource block without considering its actual communication requirement, which is spectrum inefficient. In NOMA, on the contrary, those devices are allowed to share the same resource block with other devices while strictly guaranteeing all devices' quality of service (QoS) by performing intelligent resource allocation, including power allocation, beamforming design etc. Therefore, the NOMA-based D2D communications is a more spectrally efficient framework [14]. Energy has always been a fundamental challenge for D2D devices, as many devices require continuous power and mainly rely on battery power. This leads us to use WPT to enhance energy efficiency and extend battery life. The key idea of WPT is harvesting energy from the surrounding radio frequency (RF) signals with the time-switching mode or power-splitting mode [15–19]. In this paper, only the time-switching mode is investigated.

By integrating WPT-assisted D2D communications into NOMA networks, massive connections can be realized and the aforementioned spectral and energy challenge can be effectively addressed. This paper considers integrating the WPT-assisted Inband-Underlay D2D pair [13, 20–21] into future downlink

NOMA networks. In particular, the WPT-assisted device harvests energy from the superposed downlink signals and utilizes the identical spectrum with the legacy NOMA users, where the harvested energy is used to power its transmission. Early studies provided valuable overviews, performance analyses and optimization schemes on the combination of D2D, WPT and NOMA. [22] discussed the applications, challenges and advances of WPT-enabled NOMA and WPT-assisted D2D in the future IoT. In [23], the energy efficiency of a WPT-assisted D2D pair of a downlink NOMA network was maximized, where only the single antenna scenario is considered. [24] analyzed the performance of NOMA uplink networks with multiple energy-constrained devices and one non-energy-constrained WPT device. [25] alternatively optimized the time switching coefficient and power allocation coefficients of multiple WPT-assisted devices deployed single-user downlink NOMA networks to maximize the sum rate. In [26], the energy efficiency of a downlink SWIPT-integrated NOMA system was elevated by jointly optimizing the terminals' time-switching coefficients and the power allocation of the BS. In [27], the optimal power allocation scheme for a single-uplink-user NOMA-enabled network was investigated, where only the single-carrier is used. The study incorporated a D2D transmitter and two D2D receivers and employs convex optimization techniques for determining the optimal power allocation. Another recent work by [28] enhanced energy efficiency and spectral efficiency for a WPT-assisted user-grouped uplink NOMA network, by alternating energy harvesting and signal transmission among users in these groups.

1.1 Motivations, Challenges and Contributions

Despite the extensive research already conducted in this area, most of them only considered perfect channel state information (CSI) and applied conventional optimization approaches. Against this background, this paper aims to maximize the energy efficiency of the WPT-assisted D2D transmitter deployed in multiple-input single-output (MISO)-NOMA downlink systems under imperfect CSI. However, introducing the inband-underlay D2D devices into cellular networks can lead to severe co-channel interference to those legacy downlink users due to the spectrum-

sharing protocol. Consequently, interference control plays a crucial role in preserving the QoS for existing cellular users. The presence of channel estimation errors in real-time systems also adds difficulty to the beamforming design, underscoring the importance of robust beamforming design. On the other hand, the recent successful applications of learning-based optimization methods in wireless communication, coupled with the performance degradation of traditional optimization caused by imperfect CSI, have motivated us to explore the application of learning-based methods. Furthermore, the real-time channel-dependent optimization can be viewed as the Markov decision process (MDP) and hence reinforcement learning (RL) is selected [29]. The key contributions are summarized as follows:

- This paper proposes an innovative scheme that integrates WPT-assisted D2D communications within MISO-NOMA networks. By applying the proposed scheme, spectrum and energy cooperation can be efficiently achieved. In addition, this model can be further extended to encompass the user-clustered hybrid NOMA networks [30–33]. In user-clustered hybrid NOMA, randomly distributed users have been divided into different clusters, and communication resources are allocated among clusters in an OMA manner, while users belonging to the same cluster share the same resource blocks in the NOMA manner. Thus, the proposed scheme is a promising way to realize the mMTC scenarios.
- This study investigates both convex optimization and learning-based methods to address the variable-coupled and non-concave problem of maximizing energy efficiency. For convex optimization, this paper proposes a pure fractional programming algorithm. This approach involves deriving closed-form expressions for time-switching coefficients, implementing the Dinkelbach method [34], and utilizing a complex multi-dimensional quadratic transform for beamforming design. In the context of DDPG-based solutions, this paper skillfully constructs actions, states, and the reward function. Furthermore, simulation results demonstrate the respective advantages of these two methods under varying conditions of CSI.

- The scenario of imperfect CSI is considered in this work. This paper reveals an intriguing finding: the proposed PFP algorithm demonstrates superior performance when perfect CSI can be obtained or only minor errors are present, whereas the DDPG-based algorithm showcases its ability to reduce the negative impact caused by the imperfect CSI. This finding holds substantial guidance for the practical construction and optimization of networks.
- Simulations are carried out for orthogonal multiple access (OMA) scenarios as well. The results demonstrate that the proposed PFP algorithm retains its applicability to OMA. Furthermore, simulations also show that deploying WPT-assisted D2D communication in MISO-NOMA downlink networks can provide higher energy efficiency than that in MISO-OMA downlink networks.

II. SYSTEM MODEL AND PROBLEM FORMULATION

2.1 System Model

Consider the system, where a WPT-assisted D2D pair is deployed into a legacy MISO-NOMA downlink network as shown in Figure 1. In this network, a base station (BS) with M antennas serves K single antenna NOMA downlink users (denoted by U_k). The D2D pair is permitted to utilise the identical spectrum with downlink users in the legacy NOMA network, where the D2D transmitter, denoted by D_t , is a WPT-assisted device and follows energy-harvesting-then-transmitting, and the D2D receiver is denoted by D_r . Specifically, D_t harvests energy from NOMA downlink signals during the first τT seconds and then transmits its information to D_r by using the harvested energy during the rest $(1 - \tau)T$ seconds, where $\tau \in [0, 1]$ is the time-switching coefficient and T represents one time slot duration. NOMA downlink users continuously receive signals. The integrated WPT-assisted D2D pair enable increased connectivity needs of future wireless communications in the form of efficient spectrum and energy cooperation. The introduced co-channel interference can be controlled by appropriate robust beamforming design and time coefficient allocation scheme, which constitute the primary focus of this paper. In addition, this system can be easily ap-

plied to the hybrid NOMA networks, where users are pre-clustered, as described in [30].

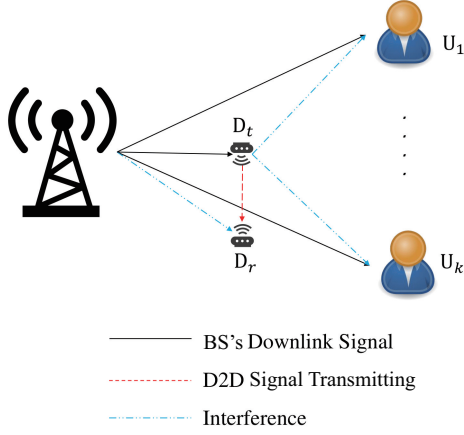


Figure 1. System model.

Denote U_k 's required signal during τ and $1 - \tau$ seconds by $s_k^{(1)}$ and $s_k^{(2)}$, satisfying $\mathbb{E}\{|s_k^{(1)}|^2\} = \mathbb{E}\{|s_k^{(2)}|^2\} = 1$, where $\mathbb{E}\{\cdot\}$ represents the expectation operator. In practical systems, CSI can not be always obtained perfectly and hence we model the channels with estimation errors in this paper. The real channels can be represented as follows [35]:

$$\hat{\mathbf{h}} = \mathbf{h} + \boldsymbol{\epsilon}, \quad (1)$$

where $\mathbf{h}$ denotes the estimated complex Gaussian channel and $\boldsymbol{\epsilon} \sim \mathcal{CN}(0, \sigma_\epsilon^2)$ symbolizes the channel estimation error.

During the first stage (i.e., τ seconds), the NOMA superposition signal $s^{(1)} = \sum_{k=1}^K \omega_k s_k^{(1)}$ is transmitted from the BS to all K downlink users and D_t performs energy harvesting from the broadcast signal $s^{(1)}$. As a result, during this stage, D_t does not cause any interference to downlink users. The received signal of k -th user during this stage can be expressed as

$$y_k^{(\tau)} = \mathbf{h}_k^H \sum_{k=1}^K \omega_k s_k^{(1)} + \boldsymbol{\epsilon}_k^H \sum_{k=1}^K \omega_k s_k^{(1)} + n_k, \quad (2)$$

where $\mathbf{h}_k \in \mathbb{C}^{M \times 1}$, $\omega_k \in \mathbb{C}^{M \times 1}$, and $\boldsymbol{\epsilon}_k$ represents the channel vector between the BS and U_k , the beamforming vector of U_k and U_k 's channel estimation error. $n_k \sim \mathcal{CN}(0, \sigma^2)$ indicates the additive White Gaussian noise (AWGN).

In this paper, the user with the weakest channel gain is denoted by U_1 , and thus U_K can be defined as the user with the strongest channel gain. This essentially means that the channel gains are arranged in ascending order, i.e., $|\mathbf{h}_1|^2 \leq |\mathbf{h}_2|^2 \leq \dots \leq |\mathbf{h}_K|^2$. Based on the principle of SIC, a stronger user is required to decode the signals of the users who have worse channel gain first. Define κ_k as the interference set when a user is decoding the k -th user's signal. Therefore, the data rate that when U_t decoding the k -th user's signal can be expressed as

$$R_{k \rightarrow t}^{(\tau)} = \tau \log(1 + \frac{|\mathbf{h}_t^H \boldsymbol{\omega}_k|^2}{\sum_{j \in \kappa_k} |\mathbf{h}_t^H \boldsymbol{\omega}_j|^2 + \Theta_t + \sigma^2}), \quad (3)$$

where Θ_t is the power of the interference caused by channel estimation errors which can be represented as $\Theta_t = \mathbb{E}\{\sum_{i=1}^K |\boldsymbol{\epsilon}_t^H \boldsymbol{\omega}_i|^2\}$. Given that $\boldsymbol{\epsilon}$ and $\boldsymbol{\omega}$ are mutually independent, and $\boldsymbol{\epsilon}_t$ values are unknown, the expression of Θ_t can be further simplified to $\Theta_t = K \sigma_\epsilon^2 \sum_{i=1}^K |\boldsymbol{\omega}_i|^2$. Once the signals from weaker users are eliminated, U_t can decode its own signal, treating the signals from other stronger users as interference. Thus, the data rate that U_t to decode its own signal of this stage is expressed as

$$R_{t \rightarrow t}^{(\tau)} = \tau \log(1 + \frac{|\mathbf{h}_t^H \boldsymbol{\omega}_t|^2}{\sum_{j \in \kappa_t} |\mathbf{h}_t^H \boldsymbol{\omega}_j|^2 + \Theta_t + \sigma^2}). \quad (4)$$

In the given MISO-NOMA downlink network, assume that D_t is equipped with a linear energy harvester and all the energy beams can be harvested [36]. Therefore, the received power at D_t can be represented as

$$P_r = \sum_{k=1}^K |\mathbf{h}_{D_t}^H \boldsymbol{\omega}_k|^2, \quad (5)$$

where $\mathbf{h}_{D_t}$ is the channel vector between the BS and D_t . Suppose that all of the energy harvested is fully utilized for signal transmission, while the energy required for the circuit is supplied by a battery. As a result, the transmission power of D_t can be calculated as follows:

$$P_t = \frac{\eta \tau P_r}{1 - \tau}, \quad (6)$$

where $\eta \in [0, 1]$ denotes the RF energy conversion coefficient.

During the rest $1 - \tau$ seconds, D_r receives the signal s_D from D_t and downlink NOMA users receive $s^{(2)}$ simultaneously on the identical spectrum. Consequently, the signal s_D , transmitted by D_t , introduces interference for U_k when it is receiving its signal $s_k^{(2)}$ and vice versa. In this context, the received signal of U_k during this stage is given by

$$y_k^{(1-\tau)} = \sqrt{P_t} h_{dk} s_D + \mathbf{h}_k^H \sum_{k=1}^K \omega_k s_k^{(2)} + \epsilon_k^H \sum_{k=1}^K \omega_k s_k^{(2)} + n_k, \quad (7)$$

where h_{dk} represent the channel from D_t and U_k . At D_r , the received signal is presented as

$$y_{D_r}^{(1-\tau)} = \sqrt{P_t} h_{dd} s_D + \sqrt{P_t} \epsilon_{dd} s_D + \mathbf{h}_{D_r}^H \sum_{k=1}^K \omega_k s_k^{(2)} + n_{D_r}, \quad (8)$$

where h_{dd} denotes the channel gain between D_t and D_r , ϵ_{dd} represents its channel estimation error, $\mathbf{h}_{D_r} \in \mathbb{C}^{M \times 1}$ represents the channel vector between D_r and the BS, and $n_D \sim \mathcal{CN}(0, \sigma_n^2)$ is the AWGN at D_r .

Considering the double-fading effect, for the downlink users, the unexpected nature of the signal transmitted by D_t , the proper decoding strategy for downlink user U_k involves directly decoding the signal from the BS while treating s_D as interference. Thus, the data rate that U_t decodes U_k 's signal is represented as

$$R_{k \rightarrow t}^{(1-\tau)} = (1 - \tau) \times \log\left(1 + \frac{|\mathbf{h}_t^H \omega_k|^2}{P_t |h_{dt}|^2 + \sum_{j \in \kappa_k} |\mathbf{h}_t^H \omega_j|^2 + \Theta_t + \sigma^2}\right). \quad (9)$$

After signals of all weaker users are removed by SIC, U_t can decode its own signal with the following rate:

$$R_{t \rightarrow t}^{(1-\tau)} = (1 - \tau) \times \log\left(1 + \frac{|\mathbf{h}_t^H \omega_t|^2}{P_t |h_{dt}|^2 + \sum_{j \in \kappa_t} |\mathbf{h}_t^H \omega_j|^2 + \Theta_t + \sigma^2}\right). \quad (10)$$

On the other hand, given the high complexity of performing SIC, it is impractical for battery-powered devices. Hence, for D_r , all signals coming from the BS are considered interference. The data rate that D_r to

decode s_D is given by

$$R_D = (1 - \tau) \log\left(1 + \frac{P_t |h_{dd}|^2}{\sum_{k=1}^K |\mathbf{h}_{D_r}^H \omega_k|^2 + P_t \sigma_\epsilon^2 + \sigma^2}\right). \quad (11)$$

Over an entire time slot, the data rate that U_k decodes its own signal and the data rate that U_t decodes other weaker users' signals are respectively given by

$$R_k = R_{k \rightarrow k}^{(\tau)} + R_{k \rightarrow k}^{(1-\tau)}, \quad (12)$$

$$R_{k \rightarrow t} = R_{k \rightarrow t}^{(\tau)} + R_{k \rightarrow t}^{(1-\tau)}. \quad (13)$$

Define P_c as the required power for driving the D_t 's circuit. The cumulative energy consumption of D_t within one time slot is given by

$$E_c = (1 - \tau) P_t + P_c = \eta \tau P_r + P_c. \quad (14)$$

2.2 Problem Formulation

The importance of effective energy utilization for WPT-enabled devices can not be overstated. Consequently, this study aims to optimize the energy efficiency of the WPT-supported D2D pair, while guaranteeing the NOMA users' QoS by jointly optimizing the BS's beamforming vectors ω_k and the time switching coefficient τ of the WPT device. The optimization problem is formulated as follows:

$$\text{P1: } \max_{\{\tau, \omega\}} \frac{R_D}{E_c} \quad (15a)$$

$$\text{s.t. } \min\{R_{k \rightarrow t}, R_{k \rightarrow k}\} \geq R_{\min}, 1 \leq k \leq t \leq K. \quad (15b)$$

$$\sum_{k=1}^K |\omega_k|^2 \leq P_{\max}, 1 \leq k \leq K. \quad (15c)$$

$$0 \leq \tau \leq 1, \quad (15d)$$

where $R_{\min}$ represents the minimum target rate of downlink users. (15b) aims to guarantee all NOMA downlink users' QoS and also to ensure that SIC can be implemented successfully. (15c) indicates the total power limitation. The constraint (15d) serves to confine the time-switching coefficient within a permissible range. $\tau = 0$ indicates that information transmitted occupied the entire time slot while $\tau = 1$ indicates D_t performs only the energy harvesting.

Lemma 1. $R_{k \rightarrow k} \geq R_{min}$ is equivalent to $\frac{1}{1-\tau} R_{k \rightarrow k}^{(1-\tau)} \geq R_{min}$, and (16) can also be recast in the same way.

Proof. Please refer to Appendix A.

Given Lemma 1, the constraint (15b) can equivalently be represented by the subsequent two constraints:

$$\frac{1}{1-\tau} R_{k \rightarrow t}^{(1-\tau)} \geq R_{min}, \quad (16)$$

$$\frac{1}{1-\tau} R_{k \rightarrow k}^{(1-\tau)} \geq R_{min}. \quad (17)$$

Specifically, (15b) can be rewrite to the following two inequations:

$$\log\left(1 + \frac{|\mathbf{h}_t^H \boldsymbol{\omega}_k|^2}{P_t |h_{dt}|^2 + \sum_{j \in \kappa_k} |\mathbf{h}_t^H \boldsymbol{\omega}_j|^2 + \Theta_t + \sigma^2}\right) \geq R_{min} \quad (18)$$

and

$$\log\left(1 + \frac{|\mathbf{h}_k^H \boldsymbol{\omega}_k|^2}{P_t |h_{dk}|^2 + \sum_{j \in \kappa_k} |\mathbf{h}_k^H \boldsymbol{\omega}_j|^2 + \Theta_k + \sigma^2}\right) \geq R_{min}. \quad (19)$$

By defining $\bar{\tau} \triangleq \frac{\tau}{1-\tau}$, (14) then can be rewritten as $E_c = \bar{\tau}(\eta P_r + P_c) + P_c$. The problem P1 then can be further simplified to

$$\text{P2: } \max_{\{\bar{\tau}, \boldsymbol{\omega}\}} \frac{\log\left(1 + \frac{\bar{\tau} \eta P_r |h_{dd}|^2}{\sum_{k=1}^K |\mathbf{h}_{D_r}^H \boldsymbol{\omega}_k|^2 + P_t \sigma_e^2 + \sigma^2}\right)}{\bar{\tau}(\eta P_r + P_c) + P_c} \quad (20a)$$

$$\text{s.t. } \frac{|\mathbf{h}_t^H \boldsymbol{\omega}_k|^2}{\bar{\tau} \eta P_r |h_{dt}|^2 + \sum_{j \in \kappa_k} |\mathbf{h}_t^H \boldsymbol{\omega}_j|^2 + \Theta_t + \sigma^2} \geq \gamma_{min}, \quad (20b)$$

$$1 \leq k < t \leq K,$$

$$\frac{|\mathbf{h}_k^H \boldsymbol{\omega}_k|^2}{\bar{\tau} \eta P_r |h_{dk}|^2 + \sum_{j \in \kappa_k} |\mathbf{h}_k^H \boldsymbol{\omega}_j|^2 + \Theta_k + \sigma^2} \geq \gamma_{min}, \quad (20c)$$

$$1 \leq k \leq K-1,$$

$$\sum_{k=1}^K |\boldsymbol{\omega}_k|^2 \leq P_{max}, 1 \leq k \leq K, \quad (20d)$$

$$\bar{\tau} \geq 0, \quad (20e)$$

where $\gamma_{min} = 2^{R_{min}} - 1$. Finding the optimal solution for P2 is challenging due to the non-concave objective function (20a) and the non-convex constraints (20b) and (20c). Additionally, the significant coupling between $\boldsymbol{\omega}$ and τ further complicates the process. The next section will discuss the way to solve P2 and provide corresponding algorithms.

III. FRACTIONAL PROGRAMMING BASED JOINT ROBUST BEAMFORMING DESIGN

This section proposes a fractional programming-based alternating algorithm to address the variable coupled non-concave problem. In particular, the problem is decomposed into two separate subproblems: the time-switching coefficient τ optimization and the robust beamforming vectors design. Each subproblem is analysed and transformed from a non-concave form to a concave form. Subsequently, convex optimization tools such as CVX or fmincon in Matlab can be applied.

3.1 Time Switching Coefficient Optimization

With the known beamforming vectors $\boldsymbol{\omega} = [\boldsymbol{\omega}_1, \dots, \boldsymbol{\omega}_K], 1 \leq k \leq K$, P2 can be recast to

$$\text{P3: } \max_{\bar{\tau}} \frac{\log\left(1 + \frac{\bar{\tau} A}{\bar{\tau} B + C}\right)}{\bar{\tau} D + E} \quad (21a)$$

$$\text{s.t. } \bar{\tau} a_t + b_{k,t} \leq 0, 1 \leq k < t \leq K, \quad (21b)$$

$$\bar{\tau} c_k + d_k \leq 0, 1 \leq k \leq K, \quad (21c)$$

$$\bar{\tau} \geq 0, \quad (21d)$$

where

$$\begin{cases} A = \eta |h_{dd}|^2 \sum_{k=1}^K |\mathbf{h}_{D_t}^H \boldsymbol{\omega}_k|^2, \\ B = \eta \sigma_e^2 \sum_{k=1}^K |\mathbf{h}_{D_t}^H \boldsymbol{\omega}_k|^2, \\ C = \sum_{k=1}^K |\mathbf{h}_{D_r}^H \boldsymbol{\omega}_k|^2 + \sigma^2, \\ D = \eta \sum_{k=1}^K |\mathbf{h}_{D_t}^H \boldsymbol{\omega}_k|^2 + P_c, \\ E = P_c, \\ a_t = \eta |h_{dt}|^2 \sum_{k=1}^K |\mathbf{h}_{D_t}^H \boldsymbol{\omega}_k|^2, \\ b_{k,t} = \sum_{j \in \kappa_k} |\mathbf{h}_t^H \boldsymbol{\omega}_j|^2 + \Theta_t + \sigma^2 - \frac{|\mathbf{h}_t^H \boldsymbol{\omega}_k|^2}{\gamma_{min}}, \\ c_k = \eta |h_{dk}|^2 \sum_{k=1}^K |\mathbf{h}_{D_t}^H \boldsymbol{\omega}_k|^2, \\ d_k = \sum_{j \in \kappa_k} |\mathbf{h}_k^H \boldsymbol{\omega}_j|^2 + \Theta_k + \sigma^2 - \frac{|\mathbf{h}_k^H \boldsymbol{\omega}_k|^2}{\gamma_{min}}. \end{cases} \quad (22)$$

Lemma 2. (21a) is always a concave-convex function in terms of $\bar{\tau}$, irrespective of the presence of channel estimation errors.

Proof. Please refer to Appendix B.

In light of Lemma 2 and the fact that the denominator of (21a) behaves as a linear function with respect to $\bar{\tau}$. It is apparent that (21a) can be categorized as a single-ratio concave-convex function. Consequently, the Dinkelbach method naturally comes to mind as a potential approach to convert it into a concave form [34].

Proposition 1. The maximum EE can be achieved when $F(q^*) = 0$, where $F(q)$ is defined as follows

$$P4: F(q) = \max_{\bar{\tau}} \log(1 + \frac{\bar{\tau}^* A}{\bar{\tau}^* B + C}) - q^*(\bar{\tau}^* D + E) \quad (23a)$$

$$s.t. \quad (21b) - (21d), \quad (23b)$$

$$\bar{\tau}^* = \min \left\{ \frac{-(AC + 2BC) + \sqrt{(AC + 2BC)^2 - 4(AB + B^2)(C^2 - \frac{AC}{D \ln 2})}}{2(AB + B^2)}, -\frac{b_{k,t}}{a_t}, -\frac{d_k}{c_k} \right\}, 1 \leq k \leq t \leq K. \quad (25)$$

Proof. Please refer to Appendix C.

3.2 Robust Beamforming Design

The previous subsection provided the optimal closed-form τ and introduced the use of the Dinkelbach method to optimize τ with given beamforming vectors ω . This section optimizes beamforming vectors under imperfect CSI with a given τ .

With a known time switching coefficient τ , P2 can be reformulated as

$$P5: \max_{\{\omega\}} \frac{\log(1 + \frac{\bar{\tau}\eta P_r |h_{dd}|^2}{P_i + \Theta_{dd} + \sigma^2})}{\bar{\tau}(\eta \sum_{k=1}^K |\mathbf{h}_{D_t}^H \omega_k|^2 + P_c) + P_c} \quad (26a)$$

$$s.t. \quad \frac{|\mathbf{h}_t^H \omega_k|^2}{\bar{\tau}\eta |h_{dt}|^2 P_r + \sum_{j \in \kappa_k} |\mathbf{h}_t^H \omega_j|^2 + \Theta_t + \sigma^2} \quad (26b)$$

where

$$q^* = \frac{\log(1 + \frac{\bar{\tau}^* A}{\bar{\tau}^* B + C})}{\bar{\tau}^* D + E}. \quad (24)$$

Proof. Please refer to [34].

By applying Proposition 1, the concavity of P3 can be established because (23a) is a concave function minus a convex function with respect to $\bar{\tau}$. Thus, the problem is simplified to a linear constraints concave problem, and hence can be solved by convex optimization tools. In the end, the optimal time switching coefficient τ^* can be obtained by $\tau^* = \frac{\bar{\tau}^*}{1 + \bar{\tau}^*}$.

Beyond the application of the Dinkelbach method, the subsequent lemma provides another efficient way to achieve the optimal $\bar{\tau}^*$.

Lemma 3. The closed-form optimal time switching coefficient can be expressed as the function of beamforming vectors ω_k when $\frac{1}{\ln 2} \frac{AC}{((A+B)\bar{\tau}+C)(B\bar{\tau}+C)} \geq D$ and $C^2 - \frac{AC}{D \ln 2} \leq 0$, which is given by the equation (25).

$$\geq \gamma_{min}, 1 \leq k < t \leq K, \quad \frac{|\mathbf{h}_k^H \omega_k|^2}{\bar{\tau}\eta |h_{dk}|^2 P_r + \sum_{j \in \kappa_k} |\mathbf{h}_k^H \omega_j|^2 + \Theta_k + \sigma^2} \quad (26c)$$

$$\geq \gamma_{min}, 1 \leq k \leq K, \quad \sum_{k=1}^K |\omega_k|^2 \leq P_{max}, 1 \leq k \leq K, \quad (26d)$$

where $P_i = \sum_{k=1}^K |\mathbf{h}_{D_r}^H \omega_k|^2$ and $\Theta_{dd} = \bar{\tau}\eta\sigma_e^2 \sum_{k=1}^K |\mathbf{h}_{D_t}^H \omega_k|^2$. The problem P5 is not a concave optimization problem because of the non-concave objective function and constraints (i.e., (26a) (26b) and (26c)). It is worth noting that the Dinkelbach method is not applicable in this case since the objective function does not take the form of a concave-convex fraction. The multidimensional and complex

quadratic transform [37] is employed to convert $P5$ into a tractable concave optimization problem.

Lemma 4. Given M functions $\mathbf{A}_m(\boldsymbol{\omega}) : \mathbb{C}^{d_1} \rightarrow \mathbb{C}^{d_2}$ and $\mathbf{B}_m(\boldsymbol{\omega}) : \mathbb{C}^{d_1} \rightarrow \mathbb{S}_{++}^{d_2 \times d_2}$, $m \in \mathbb{N}_+$, the following equivalent can be established

$$\begin{aligned} & \sum_{m=1}^M \mathbf{A}_m^H(\boldsymbol{\omega}) \mathbf{B}_m^{-1}(\boldsymbol{\omega}) \mathbf{A}_m(\boldsymbol{\omega}) \\ &= \max_z \sum_{m=1}^M (2\text{Re}\{z_m^H \mathbf{A}_m(\boldsymbol{\omega})\} - z_m^H \mathbf{B}_m(\boldsymbol{\omega}) z_m), \end{aligned} \quad (27)$$

where z_m are introduced auxiliary variables.

Proof. Please refer to Appendix D.

Proposition 2. According to Lemma 4, the objective function (26a) can be converted to the following concave form in terms of $\boldsymbol{\omega}$,

$$\begin{aligned} f_{qq}(\mathbf{W}, y, \mathbf{z}) \\ &= 2y(\log(1 + \bar{\tau}\eta|h_{dd}|^2 \sum_{k=1}^K (2\text{Re}\{z_k^H \mathbf{h}_{D_t}^H \boldsymbol{\omega}_k\} \\ &\quad - z_k^H (P_i + \Theta_{dd} + \sigma^2) z_k))^{1/2} - y^2(\bar{\tau}\eta P_r + (1 + \bar{\tau})P_c), \end{aligned} \quad (28)$$

when the following equations (30) and (29) can be satisfied respectively.

$$z_k^* = \frac{\mathbf{h}_{D_t}^H \boldsymbol{\omega}_k}{P_i + \Theta_{dd} + \sigma^2}, 1 \leq k \leq K, \quad (29)$$

$$y^* = \frac{\sqrt{R(\mathbf{W})}}{E(\mathbf{W})}, \quad (30)$$

where y and $\mathbf{z} = \{z_1, \dots, z_K\}$ are introduced auxiliary variables, $\mathbf{W}$ refers to the collection of $\{\boldsymbol{\omega}_k\}$, $R(\mathbf{W})$ is given by (E.2) and $E(\mathbf{W}) = \bar{\tau}(\eta P_r + P_c) + P_c$.

Proof. Please refer to Appendix E.

Similarly, Lemma 4 can be applied to establish the concavity of constraints (26b) and (26c). These constraints can be reformulated as two concave sets with respect to $\boldsymbol{\omega}$, as demonstrated below:

$$\begin{aligned} & \frac{|\mathbf{h}_t^H \boldsymbol{\omega}_k|^2}{\bar{\tau}\eta|h_{dt}|^2 P_r + \sum_{j \in \kappa_k} |\mathbf{h}_t^H \boldsymbol{\omega}_j|^2 + \Theta_t + \sigma^2} \\ &= \max_{\nu_{t,k}} 2\text{Re}\{\nu_{t,k}^H \mathbf{h}_t^H \boldsymbol{\omega}_k\} - \nu_{t,k}^H \alpha_{t,k} \nu_{t,k} \geq \gamma_{min}, \end{aligned} \quad (31)$$

$$\begin{aligned} & \frac{|\mathbf{h}_k^H \boldsymbol{\omega}_k|^2}{\bar{\tau}\eta|h_{dk}|^2 P_r + \sum_{j \in \kappa_k} |\mathbf{h}_k^H \boldsymbol{\omega}_j|^2 + \Theta_k + \sigma^2} \\ &= \max_{\mu_k} 2\text{Re}\{\mu_k^H \mathbf{h}_k^H \boldsymbol{\omega}_k\} - \mu_k^H \beta_k \mu_k \geq \gamma_{min}, \end{aligned} \quad (32)$$

where $\nu_{t,k}$ and μ_k are two introduced auxiliary variables whose updating functions are given by (33) and (34), $\alpha_{t,k} = |h_{dt}|^2 \bar{\tau}\eta P_r + \sum_{j \in \kappa_k} |h_t^H \boldsymbol{\omega}_j|^2 + \Theta_t + \sigma^2$ and $\beta_k = |h_{dk}|^2 \bar{\tau}\eta P_r + \sum_{j \in \kappa_k} |h_k^H \boldsymbol{\omega}_j|^2 + \Theta_k + \sigma^2$.

$$\nu_{t,k}^* = \frac{h_t^H \boldsymbol{\omega}_k}{\alpha_{t,k}}, (1 \leq t < k \leq K), \quad (33)$$

and

$$\mu_k^* = \frac{h_k^H \boldsymbol{\omega}_k}{\beta_k}, (1 \leq k \leq K). \quad (34)$$

By utilizing (28), (29), (30), (31) and (32), the problem $P5$ can be reformulated as

$$\text{P6: } \max_{\{\mathbf{W}, y, \mathbf{z}, \boldsymbol{\nu}, \boldsymbol{\mu}\}} f_{qq}(\mathbf{W}, y, \mathbf{z}) \quad (35a)$$

$$\text{s.t. } 2\text{Re}\{\nu_{t,k}^H \mathbf{h}_t^H \boldsymbol{\omega}_k\} - \nu_{t,k}^H \alpha_{t,k} \nu_{t,k} \geq \gamma_{min}, \quad (35b)$$

$$1 \leq t < k \leq K,$$

$$2\text{Re}\{\mu_k^H \mathbf{h}_k^H \boldsymbol{\omega}_k\} - \mu_k^H \beta_k \mu_k \geq \gamma_{min}, \quad (35c)$$

$$1 \leq k \leq K - 1,$$

$$\sum_{k=1}^K |\boldsymbol{\omega}_k|^2 \leq P_{max}, 1 \leq k \leq K. \quad (35d)$$

$$(30), (29), (33), (34), \quad (35e)$$

where $\boldsymbol{\nu}$ and $\boldsymbol{\mu}$ represent the collection of $\{\nu_{t,k}\}$ and $\{\mu_k\}$ respectively. Problem P6 is a concave optimization problem [38] with given y , $\mathbf{z}$, $\boldsymbol{\nu}$, and $\boldsymbol{\mu}$, since the (35a) is a concave function of $\boldsymbol{\omega}$ and constraints (35b) - (35e) are all convex set. Therefore, it can be solved by convex optimization tools. The original problem P1, which aims to maximize energy efficiency can be addressed by alternatively solving subproblems P4 and P6. Algorithm 1 summaries the proposed algorithm, namely, pure fractional programming (PFP).

Remark 1. In Algorithm 1, Step 8 (i.e., optimizing $\bar{\tau}$) can also be achieved by utilizing the Dinkelbach method. In other words, Step 8 can be replaced by “Given the optimized $\mathbf{W}$, update q by using (24) and then solve the concave problem P4 by using CVX or fmincon to obtain the optimal $\bar{\tau}^*$ ”.

Algorithm 1. *Proposed PFP algorithm.*

- 1: **Initialization:** Initialize $\mathbf{W}$ and τ to a feasible value.
 - 2: **repeat**
 - 3: Update z_k by applying (29).
 - 4: Update y by applying (30).
 - 5: Update $\nu_{t,k}$ by applying (33).
 - 6: Update μ_k by applying (34).
 - 7: Given $z_k, y, \nu_{t,k}$ and μ_k , $\mathbf{W}$ can be update by solving the concave problem P6 over $\mathbf{W}$.
 - 8: Given the optimized $\mathbf{W}$, update $\bar{\tau}^*$ according to Lemma 3 and by applying (25).
 - 9: **until** The value of (28) is convergent.
-

3.3 A Partial Exhaustive Search Based Algorithm

A partial exhaustive search-based approach to optimize ω is introduced in this subsection to verify the optimality of the proposed algorithm. This approach serves as a basis for comparison. The previous section has already demonstrated that, given τ , the original problem can be transformed into a concave problem with respect to ω . Therefore, by iterating over different τ values and solving their corresponding ω optimization problems, we can obtain all solutions to the original problem. Among them, the solution that yields the highest energy efficiency is the optimal solution. Algorithm 2 provides the detail of the partial exhaustive search approach.

Algorithm 2. *Partial exhaustive search (PES) for τ .*

- 1: **Initialization:** Initialize $\mathbf{W}$, τ and the step size ξ .
 - 2: **for** $\tau = 0.001 : \xi : 0.999$ **do**
 - 3: **repeat**
 - 4: Update z_k by applying (29).
 - 5: Update y by applying (30).
 - 6: Update $\nu_{t,k}$ by applying (33).
 - 7: Update μ_k by applying (34).
 - 8: Update $\mathbf{W}$ by solving P6, given $z_k, y, \nu_{t,k}$ and μ_k .
 - 9: **until** The value of (28) is convergent.
 - 10: **end for**
 - 11: Select the τ corresponding to the maximum (28).
-

IV. A REINFORCEMENT LEARNING BASED APPROACH

This section provides the solution of applying a reinforcement learning approach i.e., DDPG. The motivation includes: 1) the recent successful applications of reinforcement learning in wireless communications; 2) scenarios applicable to MDP; 3) performance degradation caused by imperfect CSI. In this section, we first introduce the reason for selecting DDPG, and then provide the construction and training mechanisms of the neural networks. Following that, the scheme that the application of DDPG to this specific problem is provided, including actions, states and reward setup, as well as constraints handling.

4.1 Introduction to DDPG

RL sets itself apart from other learning methods. It does not rely on labelled data sets and external supervision like supervised learning, nor does it seek to uncover hidden patterns in unlabeled data as seen in unsupervised learning [39], it learns by letting the agent interact with the environment. In the context of RL, the agent makes decisions based on the present observation, also known as the state, and consequently, it receives a corresponding reward. Broadly speaking, the objective of RL is to identify the optimal action that leads to the maximum reward. RL can typically be categorized into two types: value-based and policy-based. Q-learning and State-Action-Reward-State-Action (SARSA) are classic examples of value-based learning, but they are mainly effective for problems involving low-dimensional discrete actions. On the other hand, the policy gradient (PG), a policy-based RL approach, can handle problems involving continuous actions. But, PG tends to converge at a local optimum and does not evaluate a policy efficiently. The fusion of Q-learning and Deep Neural Networks (DNN) leads to the creation of the Deep Q Network (DQN), a technique that is apt for problems having large state spaces and discrete action spaces. For dealing with issues involving high-dimensional continuous actions, DDPG has been proposed. This method combines the strengths of DQN and PG [40]. Given that both the beamforming vectors and the time coefficient are high-dimensional continuous variables, DDPG proves to be the appropriate choice for resolv-

ing the problem discussed in this section.

4.2 Exploration and Experience Replay

Inspired by the greedy strategy, RL adds noise to the output action to encourage the agent to explore its surroundings. Following the exploration principle, the output action for the current state s_t (or observation) is given by: [40]

$$a_t = \mu(s_t|\theta_{\mu'}) + \mathcal{N}_t. \quad (36)$$

Similar to DQN, experience replay is also utilized to enhance the efficiency of the training process in DDPG [41]. To elaborate, all transitions, denoted as a, s, r, s' , are initially placed into an experience replay buffer. The training commences once this buffer is filled. Following this, a specific quantity of transitions, referred to as a mini-batch, are randomly chosen for the training of the networks.

4.3 Neural Networks: Structure and Training

Unlike other RL methods, DDPG comprises four neural networks:

- An evaluate actor network, denoted as $\mu(s|\theta_{\mu})$, which takes the current state s as input and outputs an action a .
- An evaluate critic network, represented as $Q(s, a|\theta_q)$, that inputs the current action a and state s , and outputs a Q-value.
- A target actor network, signified as $\mu'(s'|\theta_{\mu'})$, which inputs a state s' and then produces a target action a' .
- A target critic network, symbolized as $Q'(s', a'|\theta_{q'})$, which takes the target action a' and state s' as inputs and then provides the target Q-value as output.

The symbol θ stands for the parameters specific to the respective neural network. The training process can be mathematically expressed as follows.

For training the actor network, given the continuous actions and the assumption of the differentiable state-value function, the gradient ascend and chain rule are

utilized for the Q-value function

$$\nabla_{\theta_{\mu}} J = \frac{1}{N_B} \sum_{t=1}^{N_B} (\nabla_a Q(s_t, \mu(s_t|\theta_{\mu})|\theta_q) \nabla_{\theta_{\mu}} \mu(s_t|\theta_{\mu})), \quad (37)$$

where N_B represents the size of a mini-batch. The critic network is trained by minimizing the difference between the current Q-value and the target state-value:

$$L = (y - Q(s, a|\theta_q))^2, \quad (38)$$

where y is the target state-value and it is defined as:

$$y = r + \gamma Q'(s', \mu'(s'|\theta_{\mu'})|\theta_{q'}), \quad (39)$$

where γ denotes the discount factor and r represents the reward. It is important to note that the parameters θ'_{μ} and θ'_{q} of the two target networks are updated using a soft updating strategy but they are constructed with the same framework as their counterparts. The soft update can be expressed as

$$\theta' = \xi \theta + (1 - \xi) \theta', \quad (40)$$

where ξ represents the soft updating coefficient.

4.4 Application DDPG to the Problem

In this subsection, we explore the utilization of DDPG to solve the original problem P1 directly¹. Define the BS as the agent and suppose that it is capable of observing the CSI as well as the data rate of the downlink users.

- 1) Action Space: Intuitively, the variables requiring optimization, which are the beamforming and time switching coefficient should be used as the action. However, it is necessary to point out that the input vector for neural networks must be real numbers. As such, beamforming vectors are divided into their real and imaginary parts. The action taken at the t -th training step is then defined as follows:

$$a_t = [\bar{\tau}^{(t)}, \text{Re}\{\omega_1^{(t)}\}, \dots, \text{Re}\{\omega_k^{(t)}\}, \text{Im}\{\omega_1^{(t)}\}, \dots, \text{Im}\{\omega_k^{(t)}\}]. \quad (41)$$

- 2) State Space: The state vector is crafted to capture a comprehensive understanding of the current environment, as well as the effect of the action on the system. In this context of, the state vector should encompass all the CSI and the QoS requirements of all the NOMA downlink users. Therefore, the state vector at the t -th training step is determined as follows:

$$s_t = [|\mathbf{h}_1^{(t)}|^2, \dots, |\mathbf{h}_k^{(t)}|^2, |\mathbf{h}_{D_t}^{(t)}|^2, |\mathbf{h}_{D_r}^{(t)}|^2, |\mathbf{h}_{dd}^{(t)}|^2, |\mathbf{h}_{d1}^{(t)}|^2, \dots, |\mathbf{h}_{dk}^{(t)}|^2, \mathcal{R}_{t,k}^{(t)}, |\boldsymbol{\omega}_1^{(t)}|^2, \dots, |\boldsymbol{\omega}_k^{(t)}|^2], \quad (42)$$

where $\mathcal{R}_{t,k}^{(t)}, 1 \leq k \leq t \leq K$ denotes the collection of the data rate of user t to decode user k .

- 3) Reward: Our objective is to maximize energy efficiency, aligning with the aim of the DDPG algorithm which is to maximize a reward. Hence, the objective function (15a) is utilized as the positive reward when output actions can meet all constraints.

$$r_t = \left(\frac{R_D}{E_c} \right)^{(t)}. \quad (43)$$

- 4) Constraint Handling: In optimization problems, the optimized variables must comply with all constraints. When considering the application of DRL to an optimization problem, this implies that the agent's actions should be confined to a specific region. Ensuring that the output actions efficiently adhere to constraints is a significant challenge. The straightforward and harsh punitive measures is an ineffective solution. Thus, we introduce a joint approach that includes both a punishment mechanism and normalization processing to ensure that actions can satisfy all constraints. In particular, for (15a), the reward function can be rewritten as follows:

$$r_t = \begin{cases} \left(\frac{R_D}{E_c} \right)^{(t)} & \forall R_{k \rightarrow t} \geq R_{min}, \\ -\zeta |R_{k \rightarrow t}^{(t)} - R_{min}| & \exists R_{k \rightarrow t}^{(t)} < R_{min}, \end{cases} \quad (44)$$

where ζ represents the punishment factor. It's evident that if the data rate of all NOMA downlink users can reach the minimum target rate at the t -th step, the agent is rewarded positively. Conversely, if the action falls short of the target, the

agent receives a penalty. The severity of this negative consequence is determined by the extent to which the action deviates from the desired outcome. Inspired by the fact that all beamforming vectors optimized through PFP satisfy the equivalence of (15c), normalization is applied to the output beam vectors in each step. This ensures compliance with the power constraints specified in (15c). With the normalization operation, output beamforming vectors at t -th training step is given by

$$\hat{\boldsymbol{\omega}}_k^{(t)} = \sqrt{\rho_k^{(t)}} \frac{\boldsymbol{\omega}_k^{(t)}}{|\boldsymbol{\omega}_k^{(t)}|^2}, \quad (45)$$

where $\rho_k^{(t)}$ is the power allocation coefficient, which is given by

$$\rho_k^{(t)} = P_{max} \frac{|\boldsymbol{\omega}_k^{(t)}|^2}{\sum_{k=1}^K |\boldsymbol{\omega}_k^{(t)}|^2}. \quad (46)$$

As can be proved, all the output beamforming vectors in each training step always meet the total power constraint (15c) due to $\sum_{k=1}^K \hat{\boldsymbol{\omega}}_k^{(t)} = P_{max}$. On the other hand, $\hat{\boldsymbol{\omega}}_k^{(t)}$ can remain the same direction with $\boldsymbol{\omega}_k^{(t)}$. For the time switching coefficient, because $\tau \in [0, 1]$ has been converted to $\bar{\tau} \in [0, +\infty)$, we only need to map the first element of (41) to the non-negative field by using some functions, such as $\text{abs}(x)$.

Algorithm 3 presents the detailed steps of the DDPG algorithm, while the simulation section provides the structure of neural networks and corresponding parameters in detail.

Remark 2. The proposed system model can be extended to the legacy user-clustered hybrid NOMA downlink networks [30], and the proposed PFP algorithm and the DDPG-based approach are still applicable. For instance, multiple D2D transmitters and one receiver are deployed, where one transmitter is assigned to each cluster; or multiple D2D pairs are deployed, where pair is assigned to each cluster; the proposed PFP algorithm and the DDPG-based approach can be applied to each cluster.

V. SIMULATION RESULTS

In this section, the performances of the proposed algorithm and the DDPG-based algorithm are studied and compared. The same simulations are also carried out in MISO-OMA downlink networks for fair comparison.

Algorithm 3. DDPG-based algorithm.

Initialization: Initialize the four deep neural networks $\theta_q, \theta_\mu, \theta_{q'} = \theta_q$ and $\theta_{\mu'} = \theta_\mu$.
Initialize an experience replay buffer $\mathcal{D}$ with a maximum size of $\mathcal{C}$.
Set up hyperparameters $\beta_a, \beta_c, \lambda, \xi$ and N_B .
for episode $j = 1, \dots, J$ **do**
 Generate $\tau^{(j)}, \omega^{(j)}$ with random value.
 Determine the decoding order of NOMA downlink users based on their respective current channel gains
 Calculate the initial state s_1 by using (42).
 for step $t = 1, \dots, T$ **do**
 Generate a proper noise $\mathcal{N}_t$ for exploration.
 Obtain the output action by using (36).
 Retrieve beamforming vectors by using (45) and (46).
 Calculate the current states s_t .
 Calculate r_t by using (44).
 Store transition $\{s_t, a_t, r_t, s_{t+1}\}$ into $\mathcal{D}$.
 Randomly select a minibatch of N_B transitions from $\mathcal{D}$.
 Given (39) calculate the target Q value.
 Minimize the loss function (38) to update $Q(s, a|\theta_q)$.
 Update $\mu(s|\theta_\mu)$ according to (37).
 Soft update (i.e.(40)) two target networks $\theta_{q'}$ and $\theta_{\mu'}$.
 Transfer state s_t to s_{t+1} .
 end for
end for

5.1 Deep Neural Networks Parameters and Structure Setup

Both actor networks and critic networks employ fully neural networks. The actor network consists of an input layer, two hidden layers and an output layer. After the input layer and the first hidden layer, ReLU is employed as the activation function while the tanh is used

after the second hidden layer and the output layer. For the two critic networks, because their input includes state and action, the input layer and the first hidden layer are respectively divided into two individual parallel layers. Subsequently, two separated outputs are concatenated into one vector and then input to the second hidden layer. The output layer follows the second hidden layer. Only ReLU is applied in critic networks. The number of neurons is set to 500 for each hidden layer and layer normalization is also applied for efficient training. The batch size is set to $N_B = 32$. Adam optimizer is utilized with the learning rate $\beta_a = 0.001$ for actor networks and $\beta_c = 0.002$ for critic networks, and the soft update coefficient is set to $\lambda = 0.01$.

5.2 Hyper Parameters Setup

In simulations, the positions of both the D2D pair and downlink users are assumed to be randomly distributed within the region $x, y \in [3, 8]$ and BS is deployed at $(0, 0)$. Rayleigh fading channels are used in this paper, and path loss is also taken into account. The path loss coefficients between the BS and downlink users and between D_t and D_r are set to $\alpha_0 = 2.5$ and $\alpha_1 = 2$ respectively. $\alpha_2 = \alpha_3 = 3.5$ are the path loss coefficient between the BS and D_t , and between D_t and downlink users, respectively. The noise power is set to $\sigma^2 = -94$ dBm and the RF energy conversion coefficient is $\eta = 0.1$.

5.3 Simulation Results Demonstration

The proposed algorithm's convergence trends are shown in Figure 2, where the NOMA and OMA schemes are both included. It is evident that the proposed algorithm demonstrates fast convergence for both NOMA and OMA schemes, regardless of the presence of channel estimation errors. In particular, the proposed algorithm can converge within 5 iterations for all cases. Furthermore, this observation provides preliminary evidence of the superiority of NOMA over other schemes.

Figure 3 shows the relationship between energy efficiency and transmit power of the BS for different algorithms and multiple access schemes. The scenario assumes that perfect CSI can be obtained ($\sigma_e = 0$). To mitigate the impact of randomness caused by positions and channels, Monte Carlo simulations are performed to obtain averaged results. As depicted, the

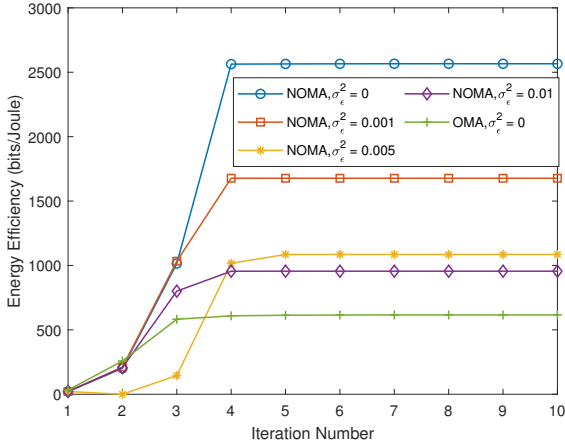


Figure 2. Convergence of the proposed algorithm, $P_{max} = 20$ dBm, $K = 3$, $M = 16$ and $R_{min} = 0.1$ bps/Hz.

PFP algorithm and the DDPG-based algorithm significantly outperform the random case, which illustrates the importance and necessity of these two proposed algorithms for this system model. For the WPT-assisted device, higher energy efficiency means that it can transmit more information with the same circuit energy consumption. The performance of the random case is such inferior due to the following two reasons: 1) the solution is feasible but not optimized; 2) D_t is not allowed to transmit when the solution cannot satisfy downlink NOMA users' QoS, and hence its energy efficiency is zero during this timeslot. On the other hand, although the PFP algorithm with the closed-form time-switching coefficient and the Dinkelbach method can provide almost the same performance, the former has incomparable advantages in complexity. The comparison between PFP and PSE shows that the alternating operation in the proposed PFP algorithm does not lose its optimality. As for the comparison between the proposed PFP and the DDPG-based, it is evident that the proposed PFP algorithm is superior to the DDPG-based algorithm when perfect CSI is available. Additionally, this figure shows that for the same algorithm and network framework, WPT-assisted D2D communication deployed in the MISO-NOMA system significantly outperforms that in the MISO-OMA system. This improvement can be attributed to the key feature of the NOMA system, which enables efficient resource sharing among users.

A significant finding of this work is illustrated in Figure 4. The proposed algorithm demonstrates supe-

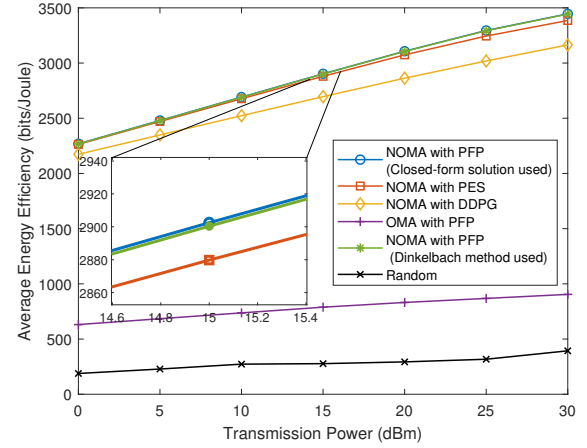


Figure 3. Energy efficiency achieved by the three algorithms as a function of the transmit power of the BS, where the obtained CSI is perfect. $K = 3$, $M = 16$ and $R_{min} = 0.1$ bps/Hz.

rior performance under conditions of perfect or near-perfect channel estimation ($\sigma_\epsilon^2 = 0$ or 0.001). However, the DDPG-based approach shows significant robustness when channel errors are severe ($\sigma_\epsilon^2 = 0.005$ or 0.01), which effectively mitigates performance degradation caused by worse channel estimation. Another notable trend the energy efficiency growth slope decreases with the increasing σ_ϵ^2 . This is due to the fact that when the BS's transmit power increases, so does the interference produced by channel estimation inaccuracy, and hence downlink users require more communication resources to maintain their QoS, which prevents the D_t from further increasing its energy efficiency.

In Figure 5, D_t 's energy efficiency in relation to the number of antennas for NOMA and OMA with different channel errors are illustrated. To distinctly highlight the differences, curves are plotted separately in three sub-figures. The most valued finding from this figure is that the NOMA scheme under the imperfect channel outperforms the OMA scheme under the perfect channel. On the other hand, A larger number of antennas results in higher energy efficiency for the D2D pair due to the benefits of spatial diversity. To improve energy efficiency, merely adding more antennas might not be an efficient scheme. As a result, when building the system, it is important to consider the trade-off between cost and performance increase.

Figure 6 demonstrates the effect of the number of

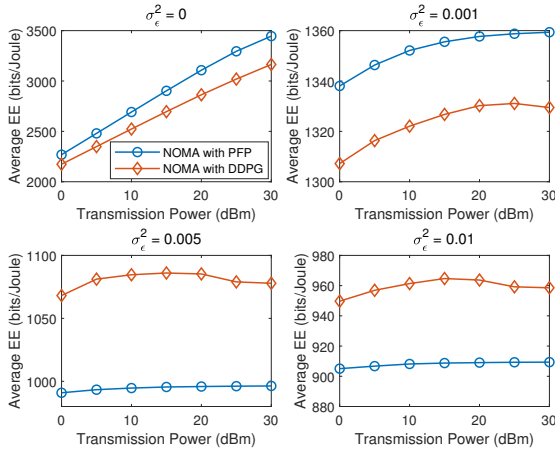


Figure 4. Impact of channel estimation accuracy to PFP and DDPG-based algorithm, where $K = 3$, $M = 16$ and $R_{min} = 0.1$ bps/Hz.

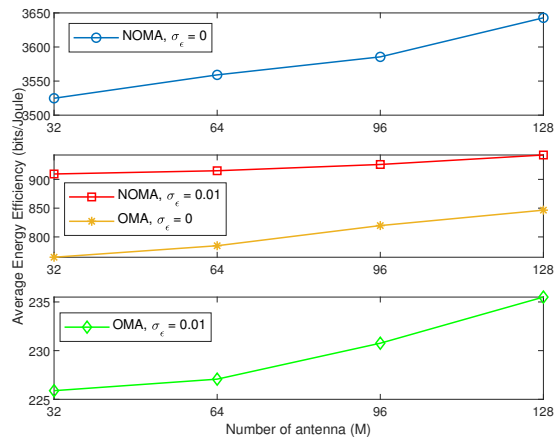


Figure 5. Energy efficiency achieved by NOMA and OMA as a function of the number of antenna equipped on the BS with different channel estimation accuracy. $K = 3$, $P_{max} = 30$ dBm and $R_{min} = 0.1$ bps/Hz.

downlink users on the energy efficiency performance of the WPT-assisted device. As seen in the figure, in the case of the NOMA scheme, with an increasing number of users, the average energy efficiency initially shows a slight increase, followed by a decrease, observed for all σ_ϵ^2 cases. It is noteworthy that for the case of $\sigma_\epsilon^2 = 0$, a slight decrease also begins when $K = 4$. However, due to the scale of the plot, this decrease is not readily apparent. Conversely, for the OMA scheme, a substantial degradation in energy efficiency occurs in all cases. The reason for this result is that in an OMA network, the increase of OMA devices

results in fewer resources allocated to each device.

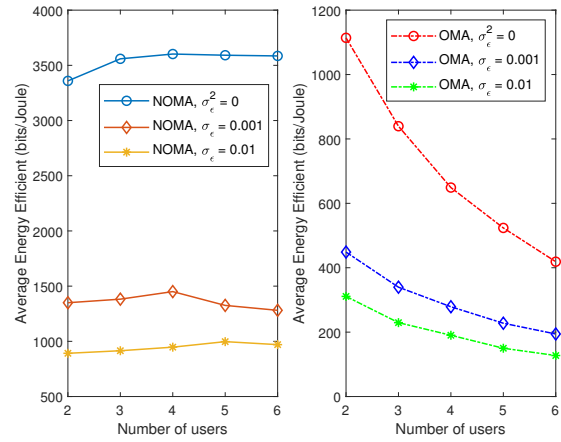


Figure 6. Energy efficiency achieved by NOMA and OMA as a function of the number of downlink NOMA users with different channel estimation accuracy, where $M = 64$, $P_{max} = 30$ dBm and $R_{min} = 0.1$ bps/Hz.

The effect of increasing QoS requirements for downlink users on the average energy efficiency of the WPT-assisted device is shown in Figure 7. The energy efficiency of WPT-assisted D2D communication in the MISO-NOMA downlink network significantly reduces as the QoS demanded by downlink users increases. Additionally, the degree of performance degradation has a positive correlation with the channel estimation accuracy. Nevertheless, the increase of OMA downlink users' QoS impacts the energy efficiency performance of D_t slightly. Particularly, the rise in the QoS demands of the OMA downlink users leads in essentially no reduction in the energy efficiency of the WPT-assisted device when the channel estimation is either perfect or has minimal errors (e.g., $\sigma_\epsilon^2 = 0$ or 0.001). Overall, although the WPT devices in the NOMA downlink network are seriously negatively affected by the QoS enhancement of downlink users, the NOMA scheme is still preferred.

VI. CONCLUSION

This paper proposed a novel scheme that can realize spectrum and energy cooperation to support the future mMTC scenarios, where imperfect CIS has been considered. The proposed signal can be simply integrated into the user-clustered hybrid NOMA network. The energy efficiency of WPT-enabled D2D de-

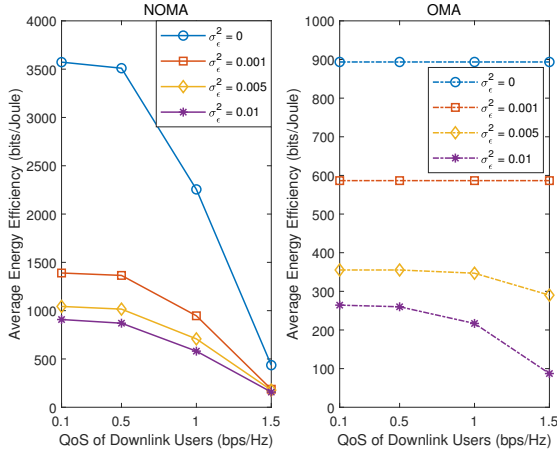


Figure 7. Energy efficiency achieved by NOMA and OMA as a function of downlink users' QoS requirement with different channel estimation accuracy, where $K = 3$, $M = 64$ and $P_{max} = 30$ dBm.

vices has been maximized by the proposed PFP algorithm and DDPG-based algorithm respectively. In the PFP algorithm, the closed-form time switching coefficient has been derived and the robust beamforming design is implemented by multidimensional and complex quadratic transform. Furthermore, to verify the PFP algorithm's optimality a partial exhaustive search based algorithm has been also proposed. On the other hand, the formulated non-concave problem has also been solved by the DDPG-based algorithm. Simulation results demonstrated the superiority of NOMA. A significant finding is that the proposed PFP algorithm outperforms the DDPG-based algorithm when channel estimation is accurate. Conversely, if channel estimation is imperfect, the DDPG-based algorithm exhibits greater robustness compared to the PFP algorithm. In the future, it will be an important direction to further explore the reasons for the above results. Furthermore, WPT with non-linear energy harvesting will be another direction. In addition, backscatter communication (BackCom) is another mature and efficient scheme to realize spectrum and energy cooperation, which will be another direction of future research.

NOTES

¹No additional processing is performed for the variables in the highly coupled non-convex problem, including the objective function and constraints.

APPENDIX

A Proof of Lemma 1

Define $\mathcal{A} = \log(1 + \frac{|h_k^H \omega_k|^2}{\sum_{j \in \kappa_k} |h_k^H \omega_j|^2 + \Theta_k + \sigma^2})$, $\mathcal{B} = \log(1 + \frac{|h_k^H \omega_k|^2}{P_t |h_{dk}|^2 + \sum_{j \in \kappa_k} |h_k^H \omega_j|^2 + \Theta_k + \sigma^2})$ and $\mathcal{R} = R_{min}$. Substitute (4) and (10) into (17). We have

$$\tau \mathcal{A} + (1 - \tau) \mathcal{B} \geq \mathcal{R}, \quad (\text{A.1})$$

which is equivalent to

$$\tau(\mathcal{A} - \mathcal{R}) \geq (\tau - 1)(\mathcal{B} - \mathcal{R}). \quad (\text{A.2})$$

Given $0 \leq \tau \leq 1$ and $\mathcal{A} \geq \mathcal{B}$, it can be proved that (A.2) can always be held if the condition $\mathcal{B} \geq \mathcal{R}$ is maintained.

B Proof of Lemma 2

- 1) For the scenario where the CSI can only be imperfectly obtained:

Define $g(\bar{\tau}) = \log(1 + \frac{\bar{\tau} A}{\bar{\tau} B + C})$ and its second-order derivative is given by

$$\frac{\partial^2 g(\bar{\tau})}{\partial^2 \bar{\tau}} = - \frac{AC(2B^2 \bar{\tau} + 2AB \bar{\tau} + AC + 2BC)}{\ln 2((A + B)\bar{\tau} + C)^2(B\bar{\tau} + C)^2}. \quad (\text{B.1})$$

As can be seen, $\frac{\partial^2 g(\bar{\tau})}{\partial^2 \bar{\tau}} \leq 0$ and hence $g(\bar{\tau})$ is a concave function of $\bar{\tau}$.

- 2) For the scenario where the CSI can be perfectly obtained:

In this case, the numerator of (21a) is reduced to $\hat{g}(\bar{\tau}) = \log(1 + \bar{\tau} \hat{A})$, where $\hat{A} = \frac{\eta P_t |h_{dd}|^2}{P_i + \sigma^2}$. The second-order derivative of $\hat{g}(\bar{\tau})$ is given by

$$\frac{\partial^2 \hat{g}(\bar{\tau})}{\partial^2 \bar{\tau}} = - \frac{\hat{A}^2}{\ln 2(1 + \hat{A})^2}. \quad (\text{B.2})$$

Given that $\hat{A}$ is also non-negative, the second-order derivative of $\hat{g}(\bar{\tau})$ is non-positive. Therefore, $g(\bar{\tau})$ and $\hat{g}(\bar{\tau})$ are both concave function of $\bar{\tau}$. The lemma is proved.

C Proof of Lemma 3

Define the numerator and the denominator of (15a) by $f_n(\bar{\tau})$ and $f_d(\bar{\tau})$ respectively, their first-order derivative can be given by

$$f'_n(\bar{\tau}) = \frac{1}{\ln 2} \frac{AC}{((A+B)\bar{\tau}+C)(B\bar{\tau}+C)}, \quad (C.1)$$

$$1 \leq k \leq t \leq K,$$

and

$$f'_d(\bar{\tau}) = D. \quad (C.2)$$

Note that (C.1) and (C.2) are both positive and hence $f_n(\bar{\tau})$ and $f_d(\bar{\tau})$ are both monotonically increasing with $\bar{\tau} \geq 0$. However, to elucidate the monotonicity of (15a), additional analysis is warranted, which will be addressed in the following discussion.

1) *Case 1:* $f'_n(\bar{\tau}) < f'_d(\bar{\tau})$.

In this case, (15a) is a monotonically decreasing function with $\bar{\tau} \geq 0$. Meanwhile, (15b) and (15c) are two increasing function of $\bar{\tau}$ since a_t and c_k are all positive. Therefore, under the premise that the constraints (15b) and (15c) can be satisfied, (15a) achieves the maximum at $\bar{\tau} = 0$ (i.e.,

$\tau = 0$). That means D_t does not perform energy harvesting, and hence this case is infeasible.

2) *Case 2:* $f'_n(\bar{\tau}) \geq f'_d(\bar{\tau})$. In this case, (15a) is a monotonically increasing function with respect to $\bar{\tau}$ and the following inequality is necessary.

$$g(\bar{\tau}) \triangleq (AB + B^2)\bar{\tau} + (AC + 2BC)\bar{\tau} + C^2 - \frac{AC}{D \ln 2} \leq 0. \quad (C.3)$$

It can be easily observed that $g(\bar{\tau})$ is a quadratic function of $\bar{\tau}$ that opens upwards, whose symmetric axis is $\bar{\tau} = -\frac{AC+2BC}{2(AB+B^2)} \leq 0$. In order to find the optimal $\bar{\tau}^*$, the following two cases need to be discussed.

- $g(\bar{\tau} = 0) = C^2 - \frac{AC}{D \ln 2} > 0$. In this case, if $g(\bar{\tau})$ does not have real roots, the region that can meet (C.3) is a null set (i.e., $T = \emptyset$). If $g(\bar{\tau})$ has real roots, the region that can meet (C.3) is given by

$$T = [x_1, x_2], \quad (C.4)$$

where x_1 and x_2 are given by (C.5).

$$x = \frac{-(AC + 2BC) \pm \sqrt{(AC + 2BC)^2 - 4(AB + B^2)(C^2 - \frac{AC}{D \ln 2})}}{2(AB + B^2)}. \quad (C.5)$$

However, the right bound of T is still negative when $g(\bar{\tau} = 0) = C^2 - \frac{AC}{D \ln 2} > 0$. Hence, the feasible region is still a null set (i.e., $T \cap [0, +\infty) = \emptyset$). Therefore, this case is also infeasible.

- $g(\bar{\tau} = 0) = C^2 - \frac{AC}{D \ln 2} \leq 0$. For this case, it can be ensured that $g(\bar{\tau})$ has two real roots. The region that can meet (C.3) is, therefore, the same as (C.4) and its right bound is positive. On the other hand, constraints (15b) and (15c) determined another feasible region, which is $[0, -\frac{b_{k,t}}{a_t}] \cap [0, -\frac{d_k}{c_k}]$. Overall, the feasible region can be derived as

$$\bar{\tau} \in [x_1, x_2] \cap [0, -\frac{b_{k,t}}{a_t}] \cap [0, -\frac{d_k}{c_k}] \cap [0, +\infty). \quad (C.6)$$

Considering the monotonicity increasing

feature of the objective function in this case, the maximum value can be achieved when $\bar{\tau}$ is located on its upper bound, i.e., $\bar{\tau}^* = \min\{x_2, -\frac{b_{k,t}}{a_t}, -\frac{d_k}{c_k}\}$.

D Proof of Lemma 4

Define $f(z_m) = 2\text{Re}\{z_m^H \mathbf{A}_m(\omega)\} - z_m^H \mathbf{B}_m(\omega) z_m$. $f(z_m)$ is a concave function in terms of z_m (a linear function minus a quadratic function yields a concave function). Furthermore, the maximum value of the concave function $f(z_m)$ can be achieved when its first-order derivative is zero (i.e., $\frac{\partial f(z_m)}{\partial z_m} = 0$). The optimal z_m^* that maximizes $f(z_m)$ can be derived as $z_m^* = \mathbf{B}_m(\omega)^{-1} \mathbf{A}_m(\omega)$. Therefore, the maximum value of $f(z_m)$ can be achieved by substitute z_m^* into $f(z_m)$ (i.e., $f(z_m^*) = \mathbf{A}_m(\omega)^H \mathbf{B}_m(\omega) \mathbf{A}_m(\omega)$). The equivalence of 4 now is established. The proof is completed.

E Proof of Proposition 2

First, we prove the equivalence between (26a) and (28). The single-ratio quadratic transform [37] is first applied to decouple the numerator and denominator of (26a) as shown below:

$$f_q(\mathbf{W}, y) = 2y\hat{R}(\mathbf{W})^{\frac{1}{2}} - y^2 E(\mathbf{W}), \quad (\text{E.1})$$

where $\hat{R}(\mathbf{W}) = \log(1 + \frac{\bar{\tau}\eta P_r |h_{dd}|^2}{P_i + \Theta_{dd} + \sigma^2})$. (E.1) is equivalent to (26a) if $f_q(\mathbf{W}, y)$ can achieve the maximum value with optimal y^* . The proof is similar to Lemma 4's proving. The first-order derivative of (E.1) with respect to y is $\frac{\partial f_q(y)}{\partial y} = 2\sqrt{\hat{R}(\mathbf{W})} - 2yE(\mathbf{W})$. Since (E.1) is a quadratic function of y , its optimal y^* can be obtained when $\frac{\partial f_q(y)}{\partial y} = 0$, which yields (30). Substitute (30) into (E.1), the objective function (26a) is retrieved and the equivalence is established. (E.1) is not a concave in terms of $\mathbf{W}$ even $-E(\mathbf{W})$ is concave due to its minus quadratic form with respect ω , because the concavity of $\hat{R}(\mathbf{W})$ can not be obtained. By applying lemma 4, the concavity of $\hat{R}(\mathbf{W})$ can be established. Thus, $\hat{R}(\mathbf{W})$ can be recast to

$$\begin{aligned} R(\mathbf{W}) = & \log(1 + \bar{\tau}\eta |h_{dd}|^2 \sum_{k=1}^K (2\text{Re}\{z_k^H h_{D_t}^H \omega_k\} \\ & - z_k^H \sum_{k=1}^K |h_{D_r}^H \omega_k|^2 z_k)). \end{aligned} \quad (\text{E.2})$$

Similarly, (E.2) is equivalent to the numerator of (26a) when (E.2) can achieve its maximum value with optimal z^* (i.e., when z_k satisfies (29)), where z denotes the collection of z_k . By combining (28), (29), (30), (E.1) and (E.2), the equivalence from (28) to (26a) is proved.

In the following, the concavity of (28) with respect to $\mathbf{W}$ is proved. Given that the second term of (28) is concave of ω due to its negative quadratic form of ω , proving the concavity of (E.2) is sufficient to show (28) is a concave function of $\mathbf{W}$.

$$\text{Define } f(\mathbf{W}) = \sum_{k=1}^K (2\text{Re}\{z_k^H h_{D_t}^H \omega_k\} -$$

$z_k^H \sum_{k=1}^K |h_{D_r}^H \omega_k|^2 z_k)$, its Hessian matrix is given by

$$\nabla^2 f(\mathbf{W}) = -2 \begin{bmatrix} z_1^{*2} \mathbf{H}_{D_r} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & z_2^{*2} \mathbf{H}_{D_r} & & \vdots \\ \vdots & & \ddots & \mathbf{0} \\ \mathbf{0} & \cdots & \mathbf{0} & z_K^{*2} \mathbf{H}_{D_r} \end{bmatrix}. \quad (\text{E.3})$$

Note that $\mathbf{H}_{D_r} = \mathbf{h}_{D_r} \mathbf{h}_{D_r}^H$ can construct a positive semidefinite matrix and $z_k^{*2} \geq 0$. Consequently, $\nabla^2 f(\mathbf{W})$ is negative semidefinite, proving $f(\mathbf{W})$ a concave function. Additionally, we know that $\log(x), x \geq 0$ is non-decreasing and concave. In accordance with the *operations that preserve convexity* for vector composition in [38], it can be demonstrated that (E.2) is concave. Similarly, considering that the function $f(x) = x^{\frac{1}{2}}$ is also concave and non-decreasing, the first term of (28) can be proved a concave function with respect to $\mathbf{W}$. Proposition 2 is proved.

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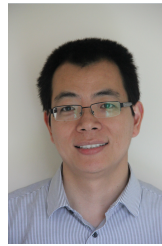
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